PROJECT SCENARIOS

Scenario 1: Hypertensive Patient Monitoring
A patient with diagnosed hypertension visits a healthcare facility where the Hypertension Prediction system is deployed. By entering their clinical parameters including blood pressure readings, symptoms, medication status, and lifestyle factors, the system provides an immediate risk assessment and classification. Healthcare providers can use this prediction to make informed decisions about treatment adjustments and patient monitoring frequency.
Scenario 2: Preventive Health Screening
A healthcare facility conducts routine health screenings for at-risk populations. Using the Hypertension Prediction system, medical staff can quickly assess individuals who may be developing hypertension before it becomes critical. The system identifies patients in early stages (Stage-1) who could benefit from lifestyle interventions and preventive measures.
Scenario 3: Emergency Department Triage
In emergency department settings, the Hypertension Prediction system assists medical staff in rapidly triaging patients with cardiovascular complaints. By analyzing vital signs and patient history, the system can identify hypertensive crises that require immediate intervention, helping prioritize critical cases.

PREREQUISITES

To complete this project, you must require the following software, concepts, and packages
Anaconda Navigator and Visual Studio:
Refer to the link below to download Anaconda Navigator

Python packages:
Open anaconda prompt as administrator
Type “pip install numpy” and click enter.
Type “pip install pandas” and click enter.
Type “pip install scikit-learn” and click enter.
Type “pip install matplotlib” and click enter.
Type “pip install scipy” and click enter.
Type “pip install pickle-mixin” and click enter.
Type “pip install seaborn” and click enter.
Type “pip install Flask” and click enter.
Prior Knowledge:
You must have prior knowledge of the following topics to complete this project.

ML Concepts
Supervised learning
Unsupervised learning
Logistic Regression
Decision tree
Random forest
Evaluation metrics
Navie Bayes
Flask Basics

PROJECT FLOW AND ACTIVITIES

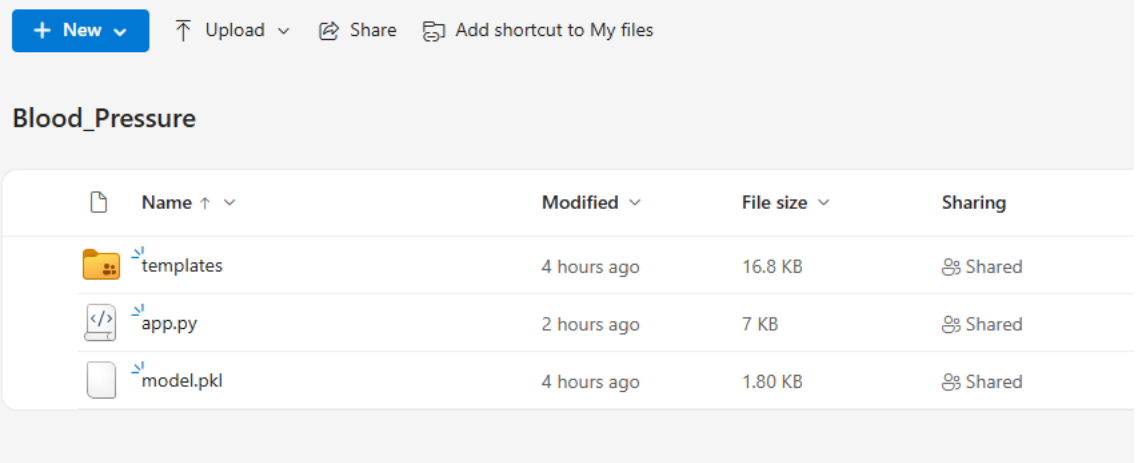
Project Flow:

- User enters patient demographic and blood test parameters through the interface
- System preprocesses the input data using the trained scaler
- Integrated machine learning model analyzes the standardized features
- Prediction result (Anemia/No Anemia) is displayed with confidence metrics
- System provides clinical interpretation and recommendations

Project Activities:
1 Data Collection & Preparation
● Collect the anemia dataset
● Data Preparation and Cleaning
2 Exploratory Data Analysis
● Descriptive statistics
● Visual Analysis
● Univariate and Bivariate Analysis
3 Model Building
● Training the model with multiple algorithms
● Testing different classifiers
4 Performance Testing & Model Selection
● Testing model with multiple evaluation metrics
● Comparing model accuracy across different algorithms
● Selecting the best performing model
5 Model Deployment
● Save the best model
● Integrate with Web Framework

PROJECT STRUCTURE

Create the Project folder which contains files as shown below



Project Structure Explanation
static/
Contains static assets (CSS, JavaScript, uploaded images)
style.css – Application styling
templates/
HTML templates for Flask application
index.html – Landing page
Flask application backend
logreg_model.pkl
Trained model file

DATASET COLLECTION AND PREPARATION

The dataset was obtained from Kaggle.com, containing 1,825 patient records with hypertension-related information.
The dataset includes the following features:
Demographic Information:
Gender (Male/Female)
Age groups (18-34, 35-50, 51-64, 65+)
Medical History:
Family history of hypertension
Current patient status
Medication usage
Time since diagnosis
Clinical Symptoms:
Symptom severity (Mild, Moderate, Severe)
Shortness of breath
Visual changes
Nosebleeds

Vital Signs:
Systolic blood pressure ranges
Diastolic blood pressure ranges
Lifestyle Factors:
Controlled diet adherence
Target Variable:
Hypertension Stages (Normal, Stage-1, Stage-2, Crisis)

Data Preparation

Data Loading and Initial Exploration
Data Cleaning Steps:
Handling Missing Values: The dataset showed no missing values after inspection
Data Type Corrections: Renamed column 'C' to 'Gender' for clarity
Inconsistency Corrections: Fixed spelling errors and standardized categorical values
Duplicate Removal: Removed 477 duplicate records from the original 1825 records

Label Encoding Applied:
Gender: Male=0, Female=1
Binary features: No=0, Yes=1
Age groups: 18-34=1, 35-50=2, 51-64=3, 65+=4
Severity: Mild=0, Moderate=1, Severe=2
Blood pressure ranges: Encoded as ordinal values
Target stages: Normal=0, Stage-1=1, Stage-2=2, Crisis=3
Feature Scaling: Applied MinMaxScaler to ordinal features for optimal model performance.

EXPLORATORY DATA ANALYSIS

1. Gender Distribution
A count plot and pie chart were used to visualize the distribution of genders in the dataset.
The count plot (Figure 1) revealed that the dataset contains an almost equal representation of male and female patients, ensuring balanced demographic coverage for analysis.
The pie chart (Figure 2) further confirmed this observation, showing near 50–50 distribution between genders. This balance minimizes gender bias during model training and interpretation.
2. Hypertension Stages Distribution

A bar chartwas used to display the count of patients across different hypertension stages.
The plot showed that Stage-1 hypertension is the most prevalent category, representing the majority of patients in the dataset. Stages 2 and 3 were less frequent, indicating fewer cases of severe hypertension. This insight suggests that most patients in the dataset are in the early phase of the condition, making early intervention analysis feasible.

3. Correlation between Systolic and Diastolic Pressure

A heatmap (Figure 4) was plotted to examine the relationship between the numeric representations of systolic and diastolic blood pressure (converted from categoricalranges to midpoints).
The heatmap displayed a strong positive correlation between systolic and diastolic values, as expected in physiological data — indicating that as systolic pressure increases, diastolic pressure also tends to rise proportionally. This validates the integrity of the recorded measurements.

4. TakeMedication vs. Severity

A count plot (Figure 5) was created to explore the relationship between medication status and hypertension severity.
The visualization showed that patients taking medication were predominantly in higher severity categories, whereas those not on medication were more frequent in lower stages or normal conditions. This trend implies a logical connection between treatment status and disease intensity.

5. Age Group vs. Hypertension Stages

Another count plot (Figure 6) was used to examine how age groups are distributed across different hypertension stages.
The analysis revealed that middle-aged and elderly individuals had a higher prevalence of Stage-1 and Stage-2 hypertension compared to younger patients. This pattern aligns with medical expectations that hypertension risk increases with age, highlighting the importance of preventive screening for older populations.

6. Pairplot: Systolic vs. Diastolic across Stages

A pairplot (Figure 7) was generated to study the multivariate relationship between systolic and diastolic blood pressure across different hypertension stages.
Distinct clusters were visible, indicating that patients with higher systolic and diastolic values tend to belong to higher hypertension stages. The distribution patterns across the diagonal density plots also confirmed increasing spread with disease severity, suggesting progressive blood pressure elevation as hypertension advances.

MODEL BUILDING

Data Splitting:
Split Results-
Training set: 1,078 sample (80%)
Testing set: 270 samples (20%)
Stratified sampling ensures balanced class representation

Algorithm and Implementation

We implemented and evaluated seven different machine learning algorithms to identify the most suitable approach for hypertension prediction:

1. Logistic Regression
Results:
Accuracy: 95.2%
Precision: High for all classes
Recall: Excellent performance across stages

2. Decision Tree Classifier
Results:
Accuracy: 100%
Performance: Perfect classification on test set
Concern: Potential overfitting indicated

3. Random Forest Classifier
Results:
Accuracy: 100%
Performance: Perfect classification
Concern: Signs of overfitting

4. Support Vector Machine (SVM)
Results:
Accuracy: 100%
Performance: Perfect classification
Concern: Potential overfitting

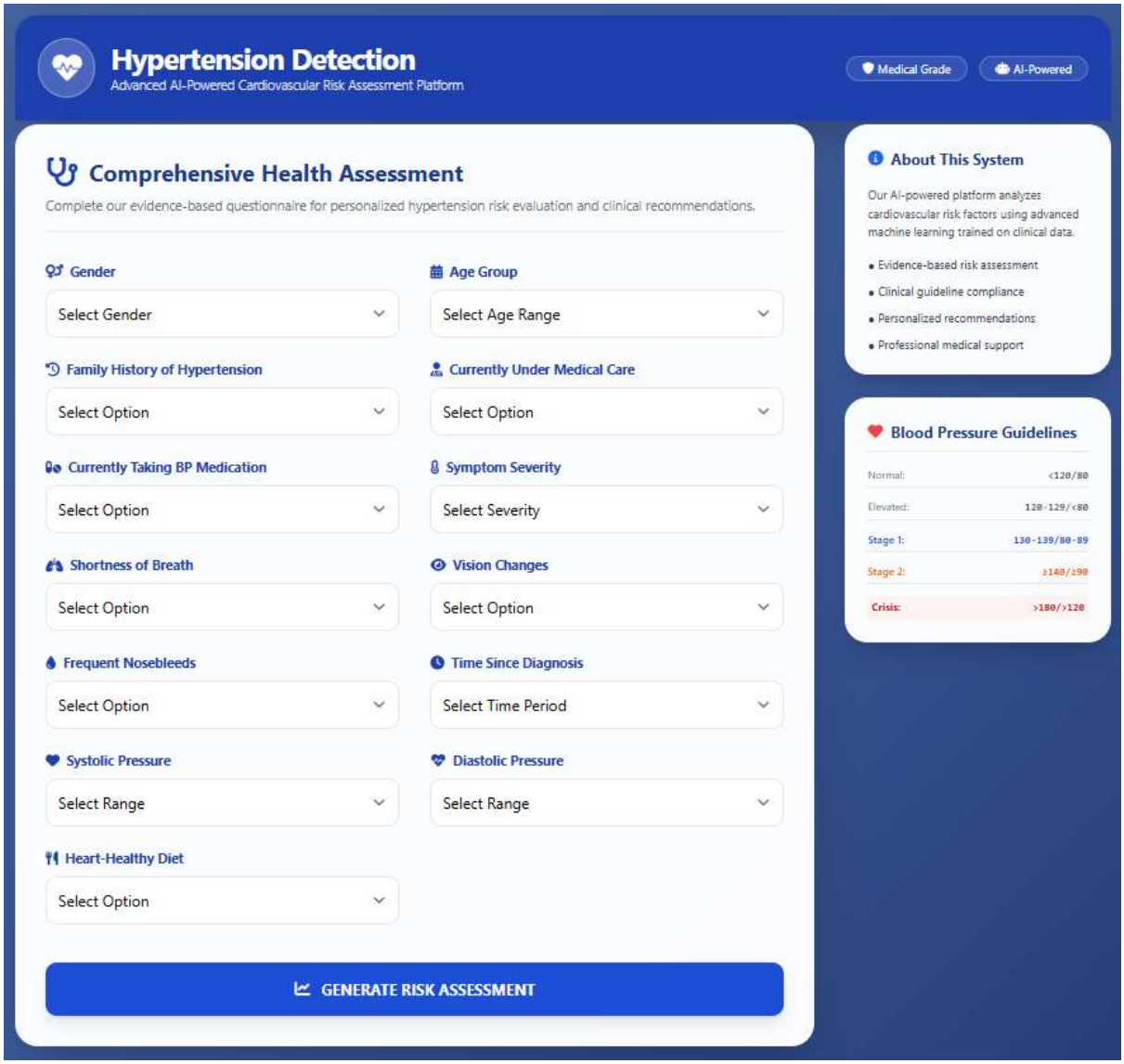
5. K-Nearest Neighbors (KNN)
Results:
Accuracy: 98.1%
Performance: Strong but not perfect
Assessment: Good generalization

6. Ridge Classifier
Results:
Accuracy: 90.0%
Performance: Solid baseline performance
Assessment: Good generalization capability

7. Gaussian Naive Bayes (Selected Model)

Model Serialization Benefits:
Persistent storage of trained parameters
Version control for model updates
Deployment-ready format
Cross-platform compatibility

MODEL DEPLOYMENT AND WEB APPLICATION



Results Include:
Hypertension stage classification
Confidence percentage
Clinical recommendations

Action items for patient care
Color-coded risk levels